

SuperStore Analysis

Chi-Squared Test

Calvin Janitra

Samuel Miduk Anugrah Pasaribu



Dataset

	🔗 order_id	🔗 order_date	🔗 ship_date	🔗 ship_mode
0	AG-2011-2040	01-01-2011	06-01-2011	Standard Class
1	IN-2011-47883	01-01-2011	08-01-2011	Standard Class
2	HU-2011-1220	01-01-2011	05-01-2011	Second Class
3	IT-2011-3647632	01-01-2011	05-01-2011	Second Class
4	IN-2011-47883	01-01-2011	08-01-2011	Standard Class
5	IN-2011-47883	01-01-2011	08-01-2011	Standard Class
6	CA-2011-1510	02-01-2011	06-01-2011	Standard Class
7	IN-2011-79397	03-01-2011	03-01-2011	Same Day
8	ID-2011-80230	03-01-2011	09-01-2011	Standard Class
9	IZ-2011-4680	03-01-2011	07-01-2011	Standard Class

51,290 rows x 21 cols

10 ▾ per page

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Chi-Squared Test

Chi-Squared Test (uji χ^2) adalah salah satu uji statistik non-parametrik yang digunakan untuk:

1. Mengukur apakah ada hubungan antara dua variabel kategorikal (contoh: Ship Mode dengan Region).
2. Mengukur apakah distribusi data sesuai dengan distribusi yang diharapkan (uji goodness-of-fit).

Jadi, intinya chi-square dipakai untuk **data kategori** (bukan numerikal).

Komponen Chi-Square Test

p-value

Probabilitas (peluang) untuk mendapatkan nilai uji sama atau lebih ekstrem dibandingkan hasil kita

- Semakin kecil p-value, semakin signifikan hubungan.
- Biasanya kita bandingkan dengan $\alpha = 0.05$.
- Kalau p-value $> 0.05 \rightarrow$ tidak signifikan (H_0 tidak ditolak).

chi-squared statistic (χ^2)

Angka hasil perhitungan dari uji Chi-Square.

- Semakin besar χ^2 , semakin besar perbedaan antara data nyata dengan yang diharapkan $\rightarrow$ indikasi ada hubungan/efek nyata.
- Kalau χ^2 kecil, berarti data nyata mirip dengan yang diharapkan $\rightarrow$ kemungkinan variabel independen (tidak ada hubungan).

degree of freedom

Jumlah informasi bebas yang bisa digunakan untuk mengestimasi variabilitas data.

df menentukan bentuk distribusi Chi-Square yang digunakan untuk menghitung probabilitas.

Uji Goodness of Fit

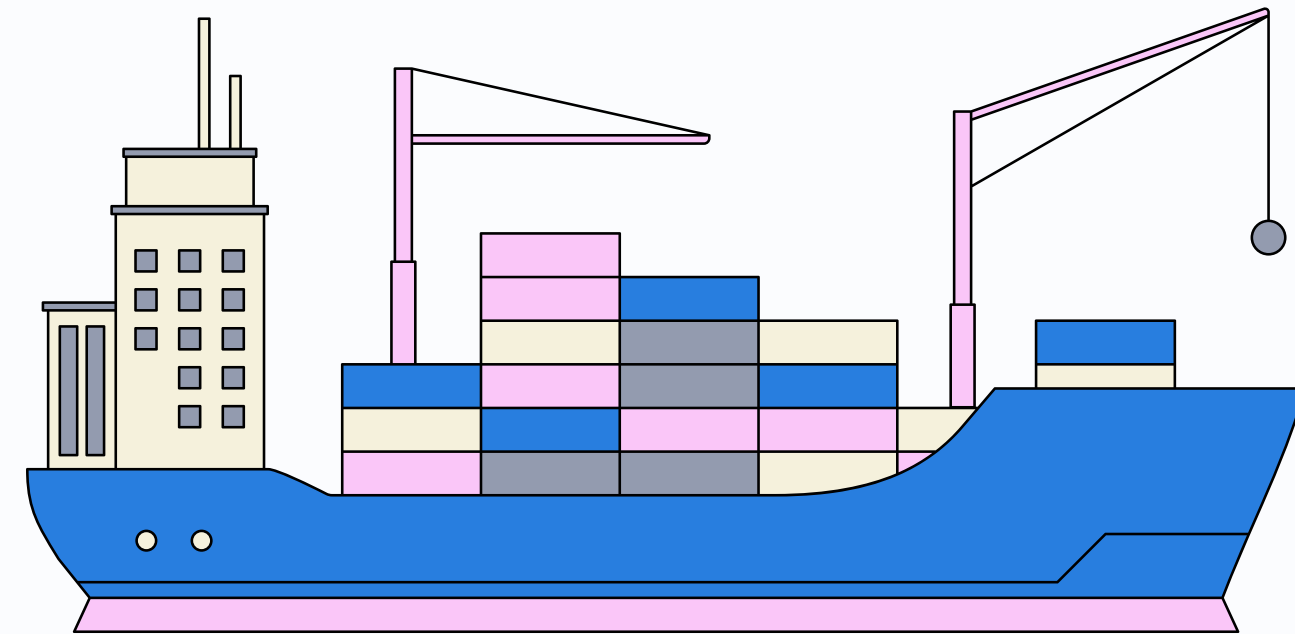
01 Ship Mode

02 Segment

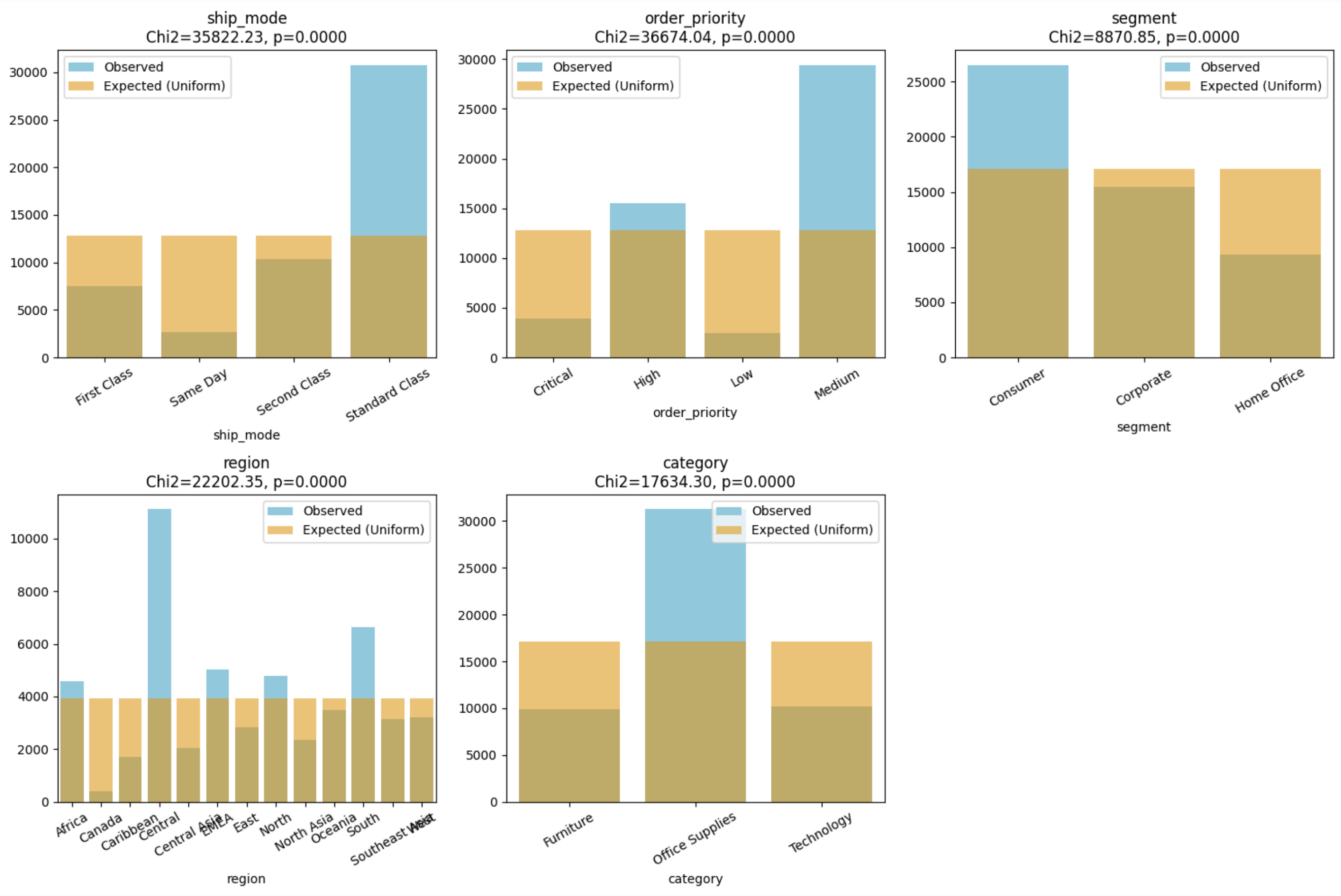
03 Region

04 Category

05 Order Priority



H₀ : Distribusi data sesuai dengan distribusi uniform, semua kategori punya frekuensi yang sama
H₁ : Distribusi data berbeda dari distribusi tidak uniform, ada kategori yang dominan atau jauh lebih sedikit



Uji Independence antar Variabel

01

Ship Mode - Order Priority

02

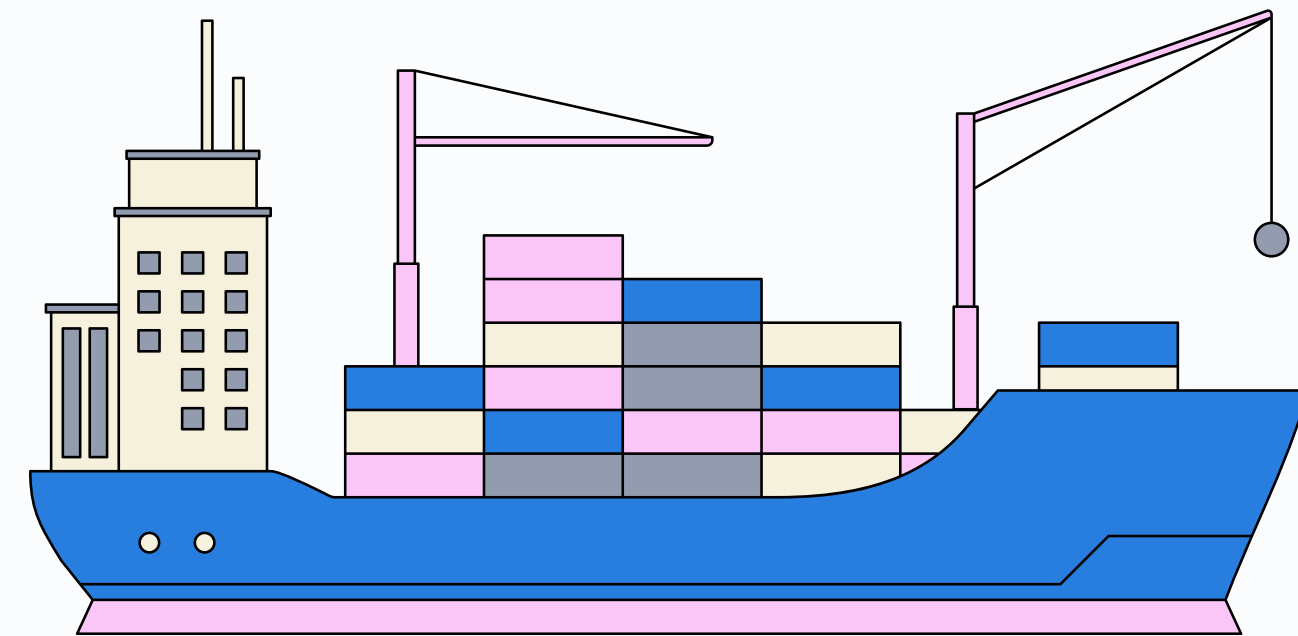
Ship Mode - Segment

03

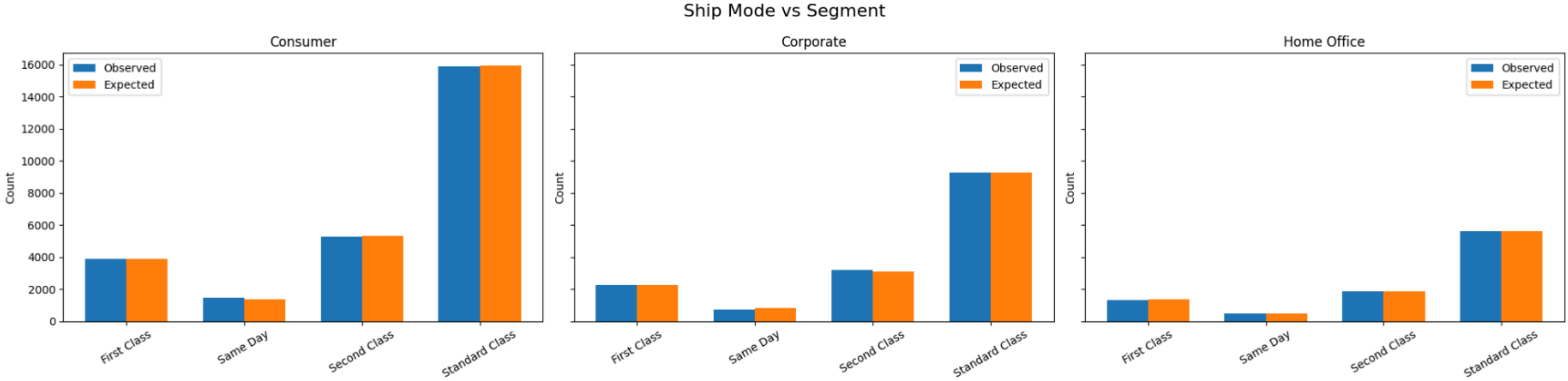
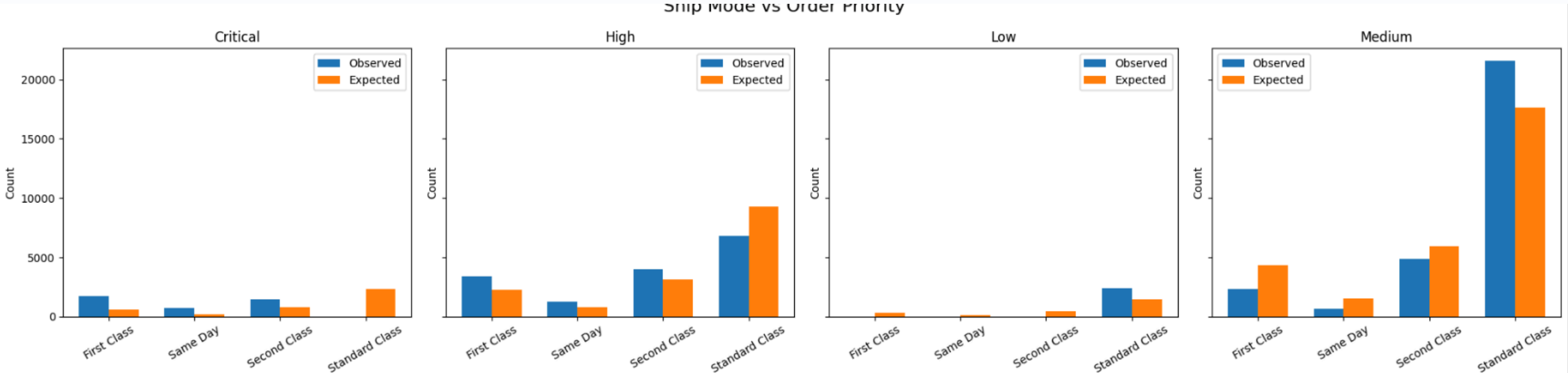
Ship Mode - Region

04

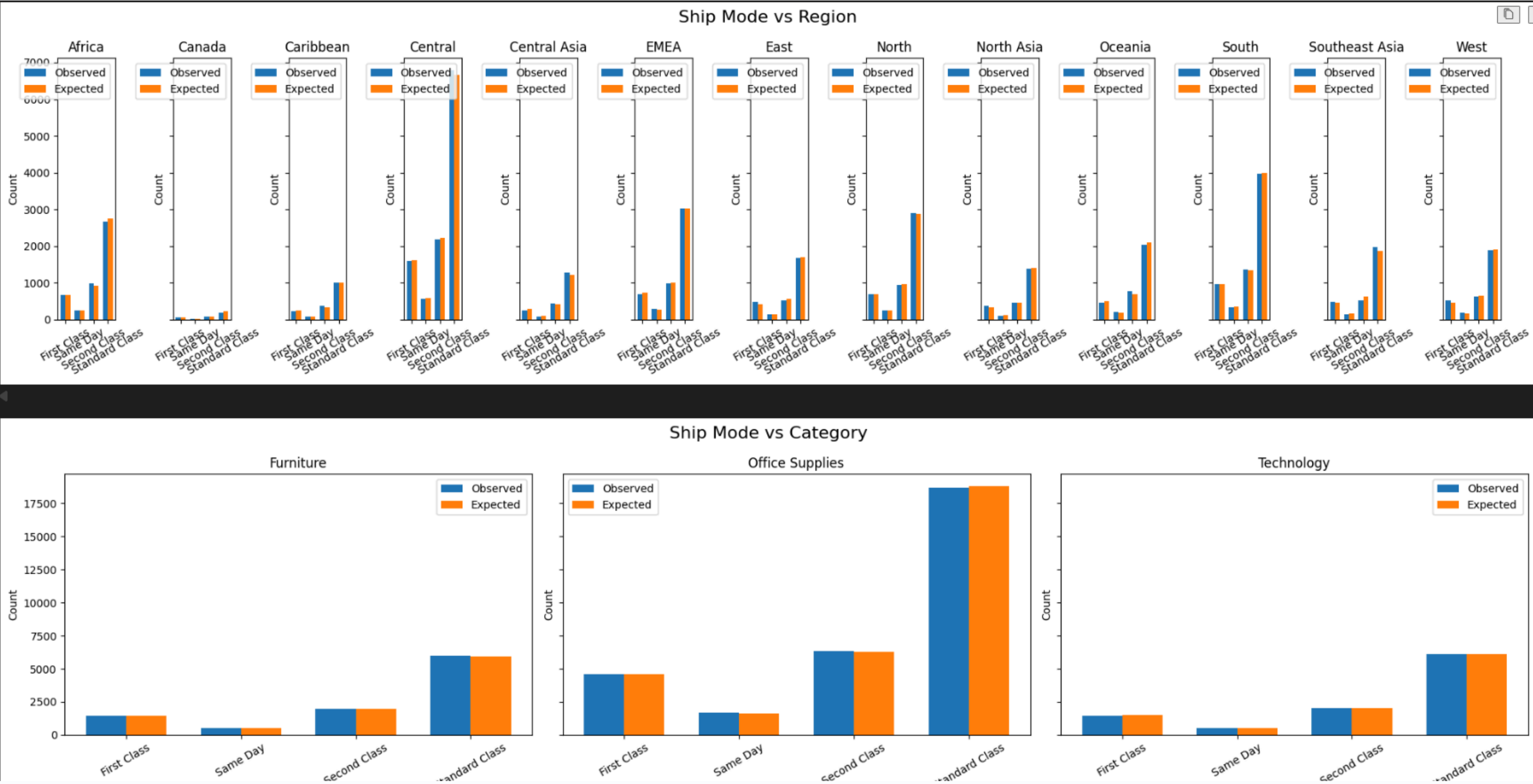
Ship Mode - Category



Observed vs Expected



Observed vs Expected



Ship Mode – Order Priority

Hipotesis

H0 : Tidak ada hubungan antara ship mode dengan order priority

H1 : Terdapat hubungan antara ship mode dengan order priority

```
=== Chi-Square Test: ship_mode vs order_priority ===
```

Contingency Table:

order_priority	Critical	High	Low	Medium
ship_mode				
First Class	1734	3413	0	2358
Same Day	742	1269	0	690
Second Class	1456	4010	0	4843
Standard Class	0	6809	2424	21542

Chi-square Statistic: 12411.07931567507

Degrees of freedom: 9

p-value: 0.0

Critical value ($\alpha=0.05$): 16.918977604620448

✓ Tolak H0 (karena 12411.08 > 16.92)

Ship Mode - Segment

Hipotesis

H0 : Tidak ada hubungan antara ship mode dengan segment

H1 : Terdapat hubungan antara ship mode dengan segment

```
=== Chi-Square Test: ship_mode vs segment ===
```

Contingency Table:

segment	Consumer	Corporate	Home Office
ship_mode			
First Class	3918	2245	1342
Same Day	1467	728	506
Second Class	5257	3198	1854
Standard Class	15876	9258	5641

Chi-square Statistic: 18.274424263215174

Degrees of freedom: 6

p-value: 0.005581770901920858

Critical value ($\alpha=0.05$): 12.591587243743977

✓ Tolak H0 (karena $18.27 > 12.59$)

Ship Mode - Region

Hipotesis

H0 : Tidak ada hubungan antara ship mode dengan region

H1 : Terdapat hubungan antara ship mode dengan region

```
=== Chi-Square Test: ship_mode vs region ===
```

Contingency Table:

region	Africa	Canada	Caribbean	Central	Central Asia	EMEA	East	\
ship_mode								
First Class	679	70	231	1593	252	700	490	
Same Day	257	29	85	568	79	300	155	
Second Class	992	91	372	2176	440	993	530	
Standard Class	2659	194	1002	6780	1277	3036	1673	

region	North	North Asia	Oceania	South	Southeast Asia	West
ship_mode						
First Class	684	385	455	958	493	515
Same Day	246	106	205	341	145	185
Second Class	954	470	779	1368	523	621
Standard Class	2901	1377	2048	3978	1968	1882

Chi-square Statistic: 140.49423628862274

Degrees of freedom: 36

p-value: 2.8283619363602042e-14

Critical value ($\alpha=0.05$): 50.99846016571065

✓ Tolak H0 (karena 140.49 > 51.00)

Ship Mode - Category

Hipotesis

H0 : Tidak ada hubungan antara ship mode dengan category

H1 : Terdapat hubungan antara ship mode dengan category

```
=== Chi-Square Test: ship_mode vs category ===
```

```
Contingency Table:
```

category	Furniture	Office Supplies	Technology
ship_mode			
First Class	1431	4598	1476
Same Day	508	1675	518
Second Class	1968	6311	2030
Standard Class	5969	18689	6117

```
Chi-square Statistic: 2.5977282981411434
```

```
Degrees of freedom: 6
```

```
p-value: 0.8573739834400929
```

```
Critical value ( $\alpha=0.05$ ): 12.591587243743977
```

```
✗ Menerima H0 (karena  $2.60 \leq 12.59$ )
```


Syarat Agar H0 Ditolak

	p-value < 0.05	χ^2 hitung > χ^2 critical
Ship Mode - Order Priority Chi Square Statistic : 12411 p-value : 0.0 df : 3		
Ship Mode - Segment Chi Square Statistic : 18.274 p-value: 0.00558 df : 6		
Ship Mode - Region Chi Square Statistic : 140.49 p-value: 2.8283619363602042e-14 df : 36		
Ship Mode - Category Chi Square Statistic : 2.5977 p-value: 0.857 df : 6		

Degrees of freedom (df)	Significance level (α)							
	.99	.975	.95	.9	.1	.05	.025	.01
1	-----	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345
4	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277
5	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086
6	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812
7	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475
8	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090
9	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666
10	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209
11	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725
12	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217
13	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688
14	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141
15	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578
16	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000
17	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409
18	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805
19	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191
20	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566
21	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932
22	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289
23	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638
24	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980
25	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314
26	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642
27	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963
28	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278
29	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588
30	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892
40	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691
50	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154
60	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379
70	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425
80	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329
100	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116
1000	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807

* significance level = 0.05

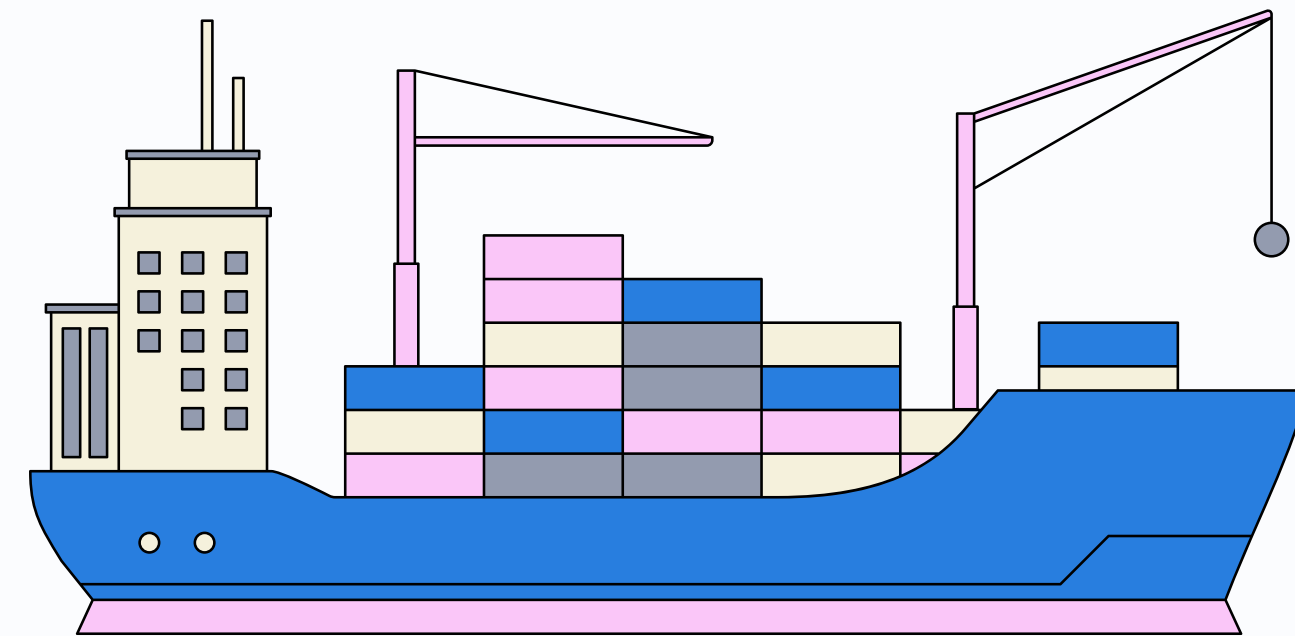
Kesimpulan

01

Distribusi variabel ship mode, segment, region, category, order priority tidak uniform

02

Variabel segment, region, order priority berhubungan dengan ship mode. Sedangkan category tidak berhubungan dengan ship mode



Thank You

